LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement #18 — Clinical Trial Subject Recruitment Engine

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1. Architecture Selection

We chose Layered Architecture for the Clinical Trial Subject Recruitment Engine.

2. Architectural Choice and Justification

The system is organized into presentation, business, and data responsibilities. The Candidate Portal and Researcher Portal handle user interaction; the Eligibility Evaluation Engine, Consent Management, and Visit Compliance components implement core business functions; and the Recruitment Database & Audit Trail stores recruitment records and audit information.

3. Reason 1 — Clear Separation of Responsibilities

The system has distinct functions: evaluating candidate survey responses against inclusion and exclusion criteria, tracking informed-consent milestones, and maintaining visit-compliance logs. Layered Architecture separates these responsibilities, making the system easier to understand, maintain, and modify without unnecessarily affecting other layers.

4. Reason 2 — Suitable for the Recruitment Workflow

The Clinical Trial Subject Recruitment Engine follows a structured flow from candidate input to eligibility evaluation and then to secure data storage. A layered design supports this predictable workflow while avoiding the additional operational and network complexity associated with a distributed microservices architecture.

5. Security Advantage

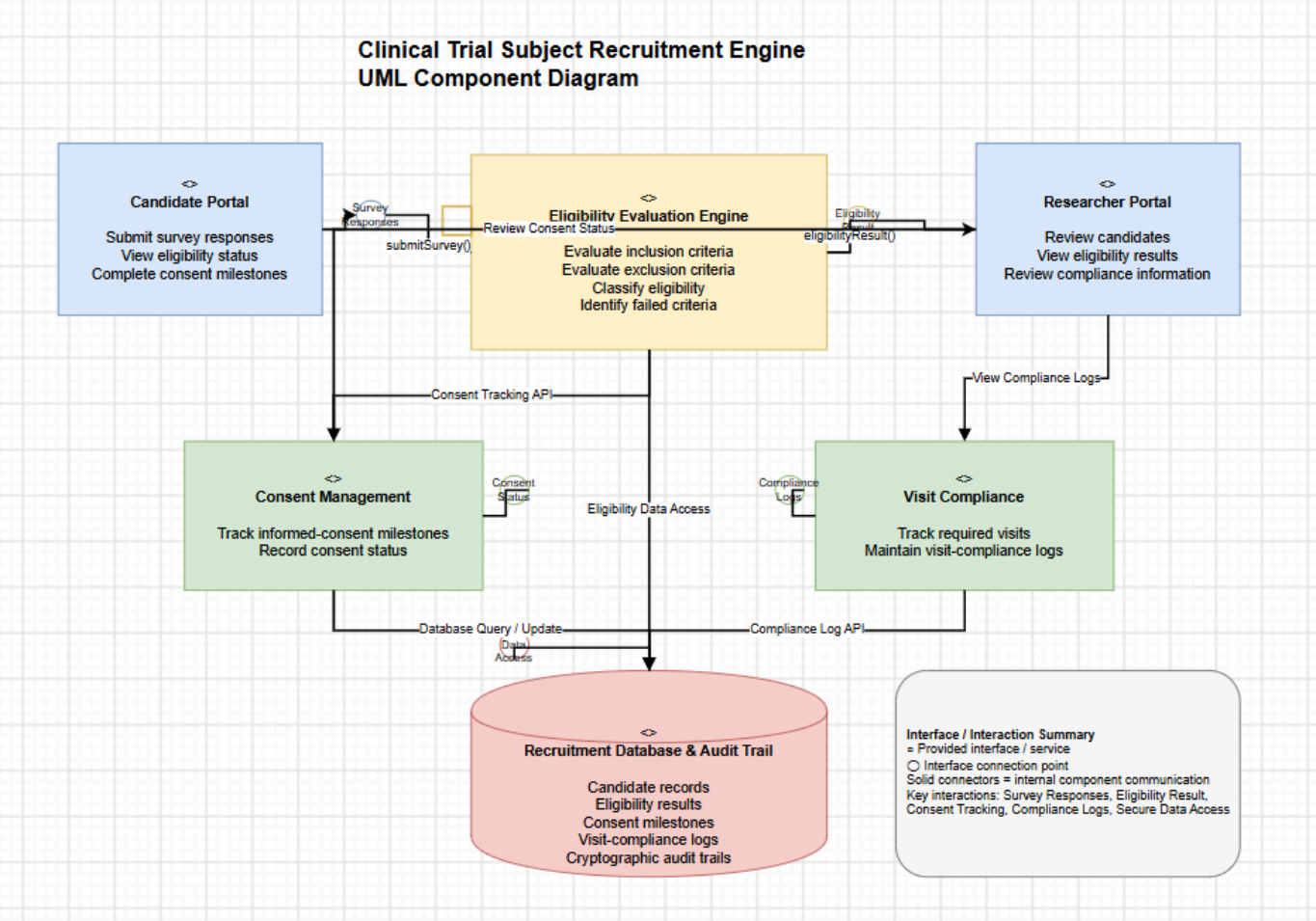
Security controls can be concentrated at the data and business layers. The Recruitment Database & Audit Trail can protect clinical-trial records and maintain the cryptographic audit trails required for 21 CFR Part 11 electronic records. Separating these responsibilities also helps control access to sensitive recruitment information and improves traceability of record changes.

6. Performance Benefit

Layered Architecture provides a direct and structured flow between the user interface, business logic, and data layer. Because the system does not require communication between numerous distributed services, it can reduce unnecessary network overhead and support efficient eligibility evaluation with acceptable latency, including under simulated peak load.

7. Components and Key Interfaces

Component	Primary responsibility
Candidate Portal	Submit survey responses, view eligibility status, complete consent milestones
Researcher Portal	Review candidates, eligibility results, and compliance information
Eligibility Evaluation Engine	Evaluate inclusion/exclusion criteria and classify eligibility
Consent Management	Track informed-consent milestones and consent status
Visit Compliance	Track required visits and maintain compliance logs
Recruitment Database & Audit Trail	Store candidate, eligibility, consent, compliance, and cryptographic audit records


Clinical Trial Subject Recruitment Engine
UML Component Diagram
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Candidate Portal
Submit survey responses
View eligibility status
Complete consent milestones
Survey Responses
submitSurvey()
<>
Eligibility Evaluation Engine
Review Consent Status
Evaluate inclusion criteria
Evaluate exclusion criteria
Classify eligibility
Identify failed criteria
Eligibility Result
eligibilityResult()
<>
Researcher Portal
Review candidates
View eligibility results
Review compliance information
View Compliance Logs
Consent Tracking API
<>
Consent Management
Track informed-consent milestones
Record consent status
Consent Status
Eligibility Data Access
Compliance Logs
<>
Visit Compliance
Track required visits
Maintain visit-compliance logs
Database Query / Update
Data Access
Compliance Log API
<>
Recruitment Database & Audit Trail
Candidate records
Eligibility results
Consent milestones
Visit-compliance logs
Cryptographic audit trails
Interface / Interaction Summary
= Provided interface / service
Interface connection point
Solid connectors = internal component communication
Key interactions: Survey Responses, Eligibility Result, Consent Tracking, Compliance Logs, Secure Data Access